SPiiPlus Utilities

User Guide

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ACS MOTION CONTROL

SPiiPlus Utilities

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PATENTS

Israel Patent No. 235022
US Patent Application No. 14/532,023
Europe Patent application No.15187586.1
Japan Patent Application No.: 2015-193179
Chinese Patent Application No.: 201510639732.X
Taiwan(R.O.C.) Patent Application No. 104132118
Korean Patent Application No. 10-2015-0137612

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Revision History

Date	Revision	Description
December 2023	3.14	New Remote Connections UI
June 2023	3.13.01.01	Blank pages fixed
February 2023	3.13.01	New Release
November 2022	3.13	New Release
April 2022	3.12	Corrections to UMD simulator, added images
November 2021	3.11.01	New Release
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June 2020	3.00	Version 3.00
June 2017	2.40	Reformatted with new template for SPiiPlus ADK Suite v2.40
August 2016	2.30	Updated for SPiiPlus ADK Suite v2.30 Added Utilities guide extension section for the SPiiPlus simulator
March 2015	2.29	First release

Conventions Used in this Guide

Text Formats

Format	Description
Bold	Names of GUI objects or commands
BOLD + UPPERCASE	ACSPL+ variables and commands
`Monospace + grey background`	Code example
Italic	Names of other documents
Blue	Hyperlink
[ ]	In commands indicates optional item(s)
|	In commands indicates either/or items

Flagged Text

	**Note** - includes additional information or programming tips.
	**Caution** - describes a condition that may result in damage to equipment.
	**Warning** - describes a condition that may result in serious bodily injury or death.
	**Model** - highlights a specification, procedure, condition, or statement that depends on the product model
	**Advanced** - indicates a topic for advanced users.

Related Documents

Documents listed in the following table provide additional information related to this document.

The most updated version of the documents can be downloaded by authorized users from ACS Downloads.

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1. Document Scope

The information in this document describes the following utilities:

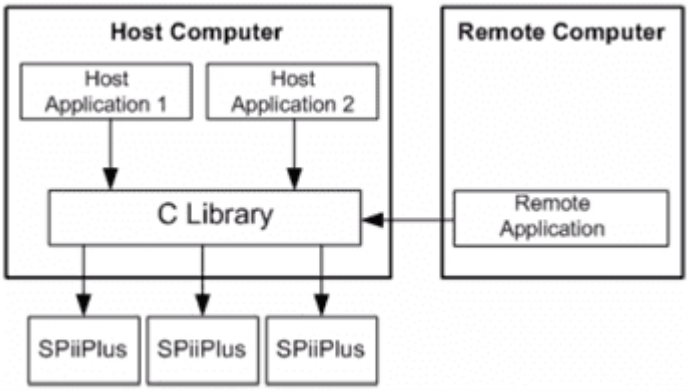
- The SPiiPlus User Mode Driver is the communication link between the host computer and the SPiiPlus Motion Controller and SPiiPlus Simulator.
- The SPiiPlus Upgrader (**UPGRADER.EXE**) is an executable program to upgrade a controller's firmware.
- The SPiiPlus Simulator is a simulator for application development and debugging.

2. SPiiPlus User Mode Driver

The SPiiPlus User Mode Driver is the communication link between the host computer and the SPiiPlus Motion Controller and the SPiiPlus Simulator. This section gives the procedures to use the User Mode Driver.

2.1 SPiiPlus User Mode Driver Overview

The SPiiPlus User Mode Driver is the communication link between applications running on the host computer and the SPiiPlus Motion Controllers in the system (or the SPiiPlus Simulator). It is also the communication link, via the host computer, between applications running on remote computers.



2.2 To View the SPiiPlus User Mode Driver

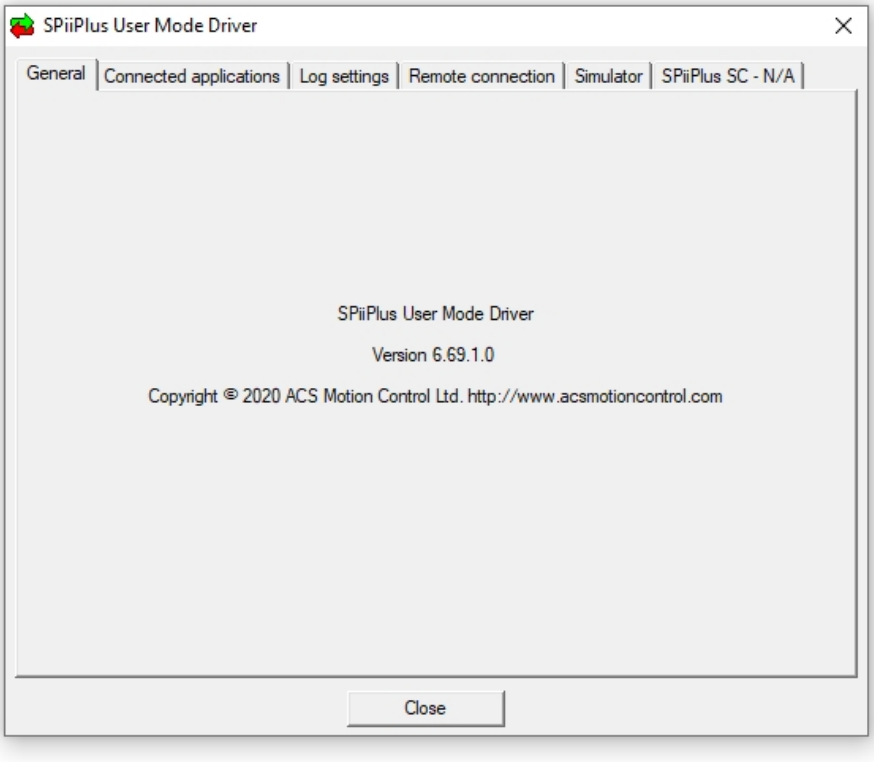
The SPiiPlus User Mode Driver has five functions:

1. Handling connected applications
2. Establishing log settings
3. Establishing a remote connections
4. Running SPiiPlus simulator
5. Establishing SPiiPlusSC settings

The SPiiPlus User Mode Driver is in the Windows start-up tray:



1. To display the SPiiPlus User Mode Driver, **double-click** the icon in the tray:



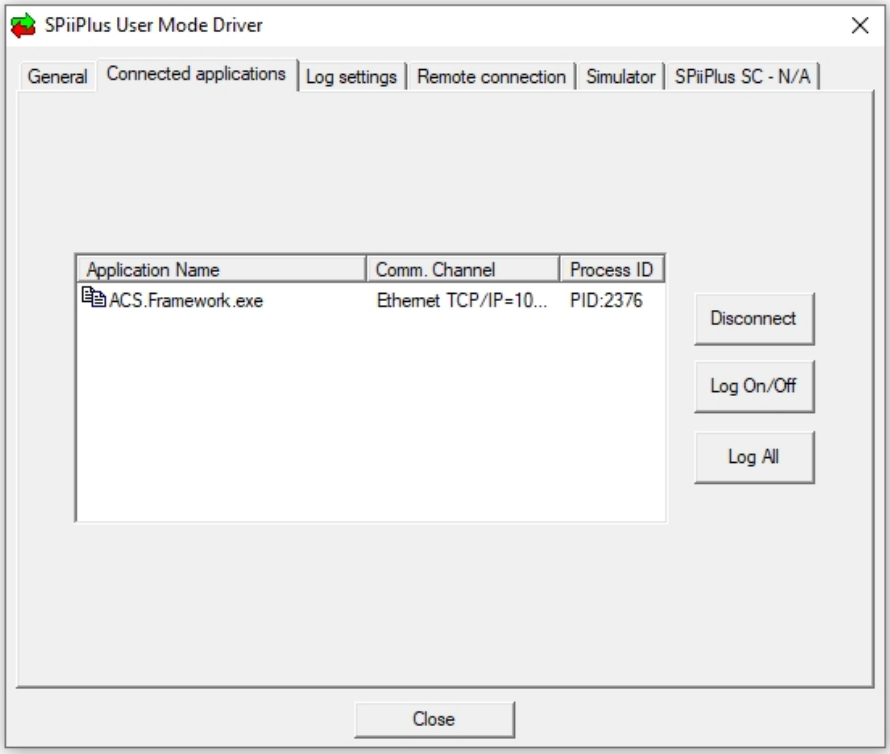
An alternate procedure to display the SPiiPlus User Mode Driver is to r**right-click** the icon and selecti **Open**.

2. To close the window click **Close**.

2.3 Handling Connected Applications

To view applications connected to the SPiiPlus Motion Controller:

1. Display the SPiiPlus User Mode Driver window.
2. Click the Connected applications tab:



All applications currently running and connected to the motion controllers are listed by:

- **Application Name** - the path and name of the application
- **Comm. Channel** - the channel through which communication with the controller is established, and may be:
  - Simulator - the application is connected to the SPiiPlus Simulator
  - Serial - the application is connected to the controller via serial communication
  - Ethernet - the application is connected to the controller via Ethernet communication
- **Process ID** - the application's process ID

Options are:

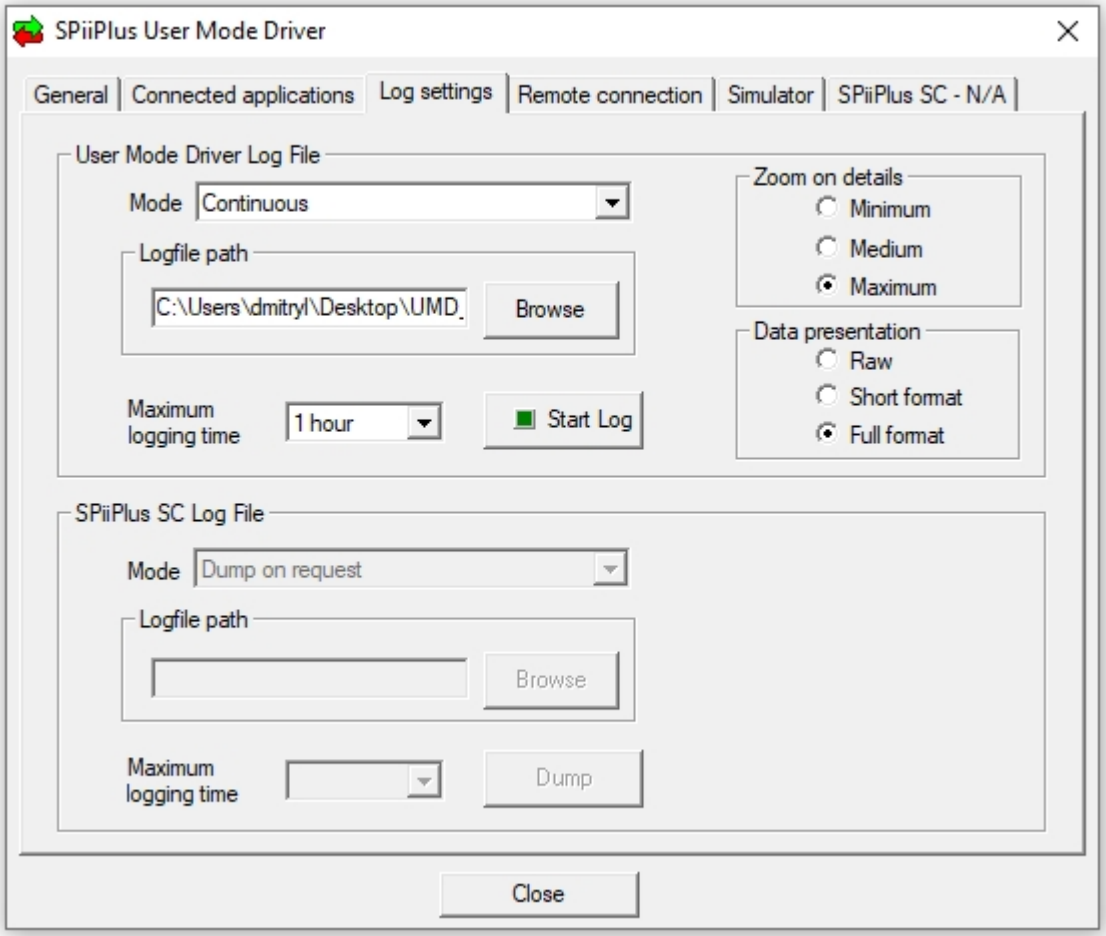
- **Disconnect** - disconnects the selected application from the SPiiPlus Motion Controller to which it is connected.
- **Log On/Off** - toggles the logging of the selected application between On and Off.
- **Log All** - causes all applications to be logged.

2.4 Log Handling

The SPiiPlus User Mode Driver constantly logs at runtime. The data is stored in binary format in an internal cyclic buffer and is translated to text just before it is written to file.

To access the log options:

1. Display the SPiiPlus User Mode Driver window.
2. Click the **Log settings** tab:



There are two types of log files:

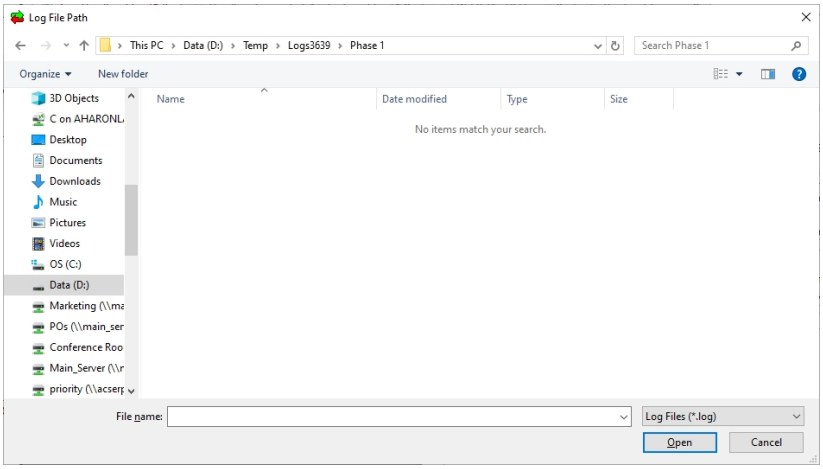
- **User Mode Drive Log File** - This log file contains the User Mode Driver activity.
- **SPiiPlusSC Log File** - This log file contains the SPiiPlusSC activity detected by the User Mode Driver.

SPiiPlusSC shared memory updates are not logged by User Mode Driver. However, the Windows host application can add shared memory updates to the log file using the functions:

- **acsc_WriteLogFile**
- **acsc_WriteSCLogFile**

The User Mode Driver initially displays the default log file name and location. You have the option of changing this by:

1. Click **Browse**, this displays the **Log File Path** window:



2. Use the browser to establish the path in the **Look in** field, and, if you desire, enter a different file name in the the **File name** field.
3. Click **Open**.

You can select the amount of details and how the data is formatted by selecting the appropriate radio button in **Zoom on details** and **Data presentation**.

Logging has two modes (selected from the **Mode** dropdown menu):

2.4.1 Log Dump on Request

In this mode the log data is saved to a file when you click **DUMP**.

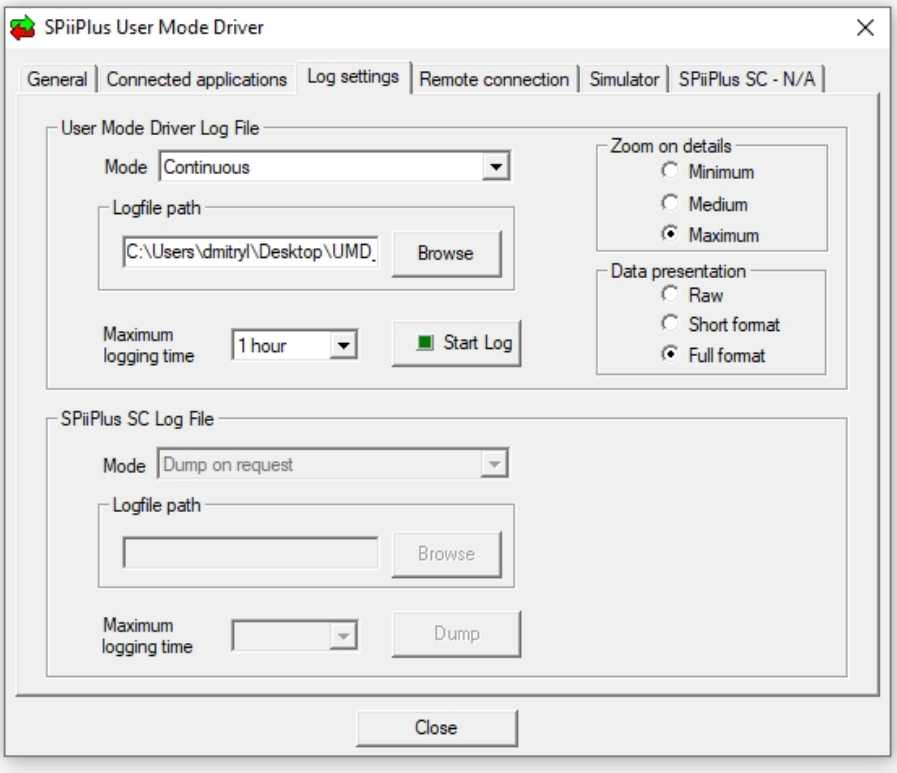


This "Log Dump on Request" is the same operation as calling the C Library function: **acsc_FlushLogFile** (see *SPiiPlus C Library Reference Guide*).

2.4.2 Continuous Log

The **Continuous** mode is a background thread that takes care of periodic file updates. It reads the binary buffer and performs text formatting to a specified file.

When you select **Continuous** from the **Mode** dropdown list, the window changes to:



1. Select the maximum time of logging from the **Maximum logging time** dropdown list. The time ranges from hours, to days, to Infinite (that is, it always runs).

The **Infinite** setting is not recommended because of a possible disk overflow situation.

2. To start logging click **Start Log**.

The button changes to **Stop Log**. when you click this button, logging stops.

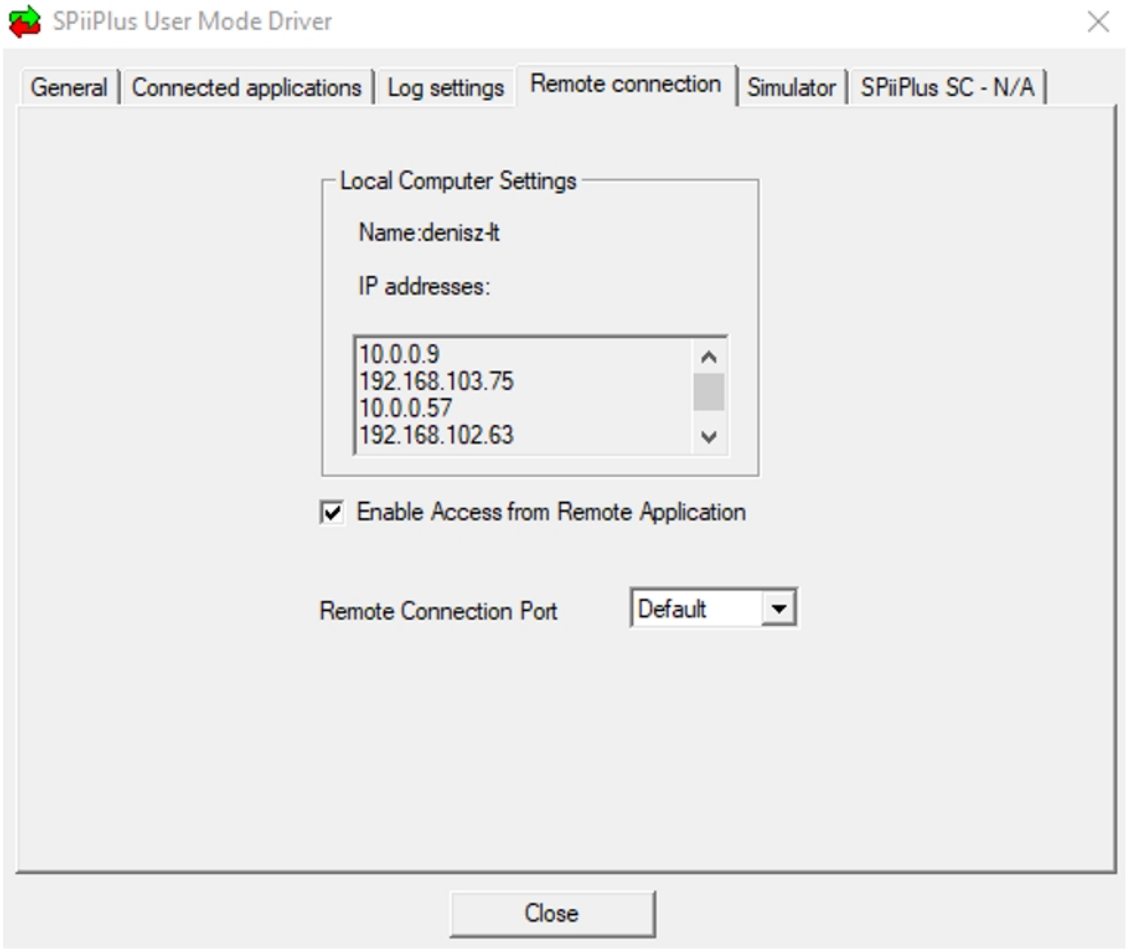
2.5 Remote Connections

Prior to establishing communication with a SPiiPlus controller or the SPiiPlus Simulator, remote applications should use the SPiiPlus C Library function: **acsc_SetServerExtLogin** (see the *SPiiPlus C Library Reference Guide*).

2.5.1 Setting Up Remote Connection

To set up a remote connection:

1. Display the SPiiPlus User Mode Driver window.
2. Click the **Remote connection** tab:



If the default port is not busy, no communication error messages are encountered, and no problem is anticipated. In this case, use the C Library **acsc_SetServerExtLogin** function with the ACSC_DEFAULT_REMOTE_PORT parameter to set the remote port address from the user application (for details on the command see *SPiiPlus C Library Reference Guide*).

If the default port (9999) is busy, the User Mode Driver returns the following error message:

> "Requested port 9999 is in use by another application. Select another port in the Remote Connection tab."

If you receive this message, proceed as follows:

1. Select **Change** from the **Remote Port Connection** drop down list. The **Enter valid port number** prompt is displayed:



2. Enter the remote port address and click **OK**.
3. Select the **Enable Access from Remote Application** checkbox. As soon as the check box is selected, communication with the remote port is attempted. If communication does not succeed, the following error message appears:

   "Requested port [port number] is in use by another application. Select another port in the Remote Connection tab."
4. Repeat Steps 1-3 until communications are established. When communications are successfully established, the User Mode Driver stores the settings.
5. Click **Close**.
6. In the user application, use the C Library **acsc_SetServerExtLogin** function and specify the same port number that was entered in the User Mode Driver.

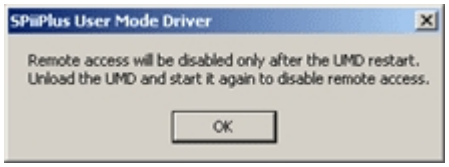


Every time the User Mode Driver initializes, the availability of the specified port is checked. If the system configuration or port number have changed, the User Mode Driver generates an error message and the **Enable Access from Remote Application** check box is cleared.

2.5.2 Disabling Remote Access

To disable the remote access:

1. Display the SPiiPlus User Mode Driver window.
2. Click the **Remote connection** tab.
3. Deselect the **Enable Access from Remote Application** checkbox and click **Close**. A User Mode Driver prompt appears.



Unload the User Mode Driver and then restart it as detailed in Unloading and Reloading SPiiPlus User Mode Driver.

2.6 SPiiPlus Simulator

Use the SPiiPlus User Mode Driver to setup the SPiiPlus Simulator. After setup, the SPiiPlus Simulator is run from the following interfaces:

> User Mode Driver

- MMI Application
- Studio SPiiPlus C Library
- SPiiPlus Com Library
- SPiiPlus .NET Library



See SPiiPlus Simulator for instructions for running the SPiiPlus Simulator using one of the interfaces.

Activate the Simulator using the SPiiPlus User Mode Driver:

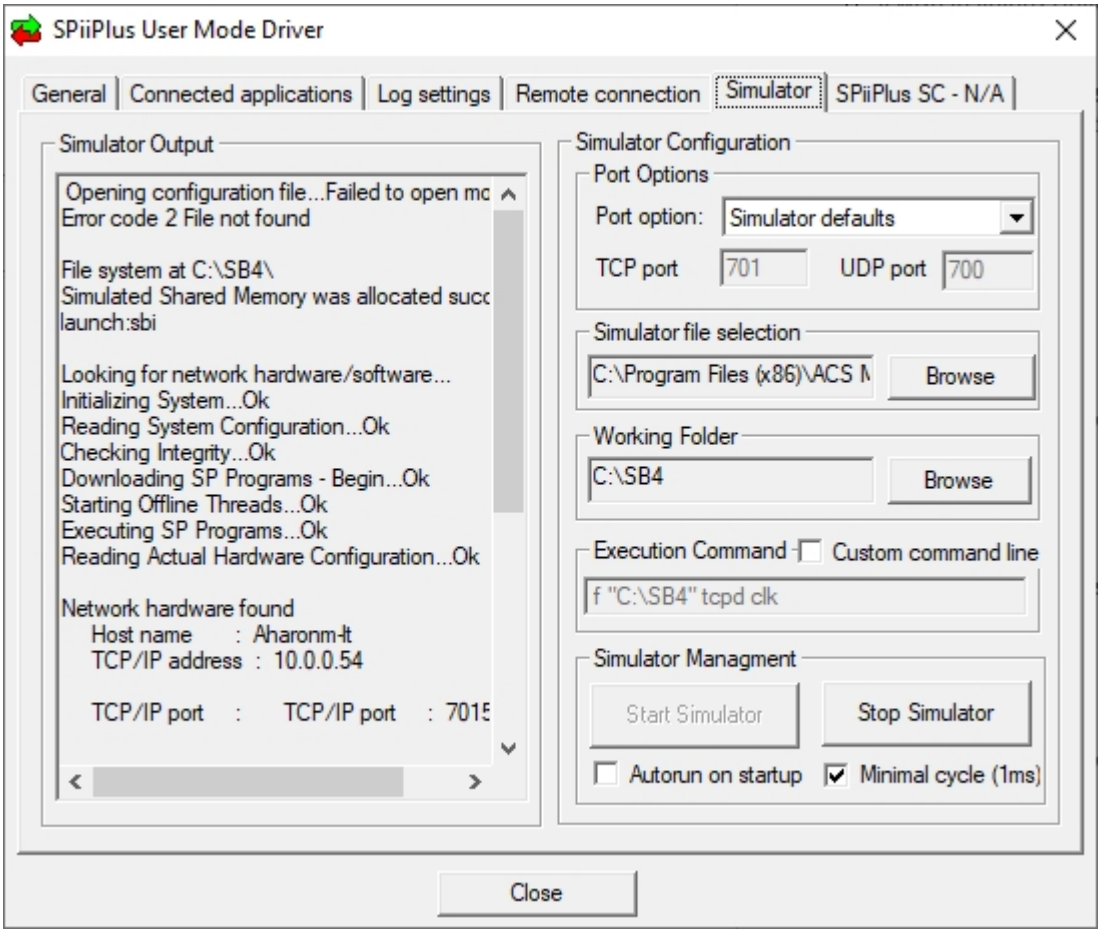
1. Display the SPiiPlus User Mode Driver window, see To View the SPiiPlus User Mode Driver
2. Click the **Simulator** tab.

Setting up the User Mode Driver to run the SPiiPlus simulator:

1. Selecting a SPiiPlus simulator executable file.
   a. Click **Browse** in the Execution Options pane.
   b. Select the simulator executable file.
2. Selecting the Working Folder in which the simulator will be executed.
   a. Click **Browse** in the Working Folder pane.
   b. Select execution location.
3. Changing the TCP port.
   a. Click the down arrow in the Port option field.
   b. Select **User specific**.
   c. Enter port number.
4. Changing the UDP port.
   a. Click the down arrow in the Port option field.
   b. Select **User specific**.
   c. Enter port number.
5. (Optional) uncheck the **Minimal cycle**. This option will affect the Windows scheduling.
6. (Optional) check the **Autorun on startup**. If selected, the simulator will automatically be executed whenever the User Mode Driver is started.



The parameters input into the setup procedure are stored for future executions of the SPiiPlus User Mode Driver.



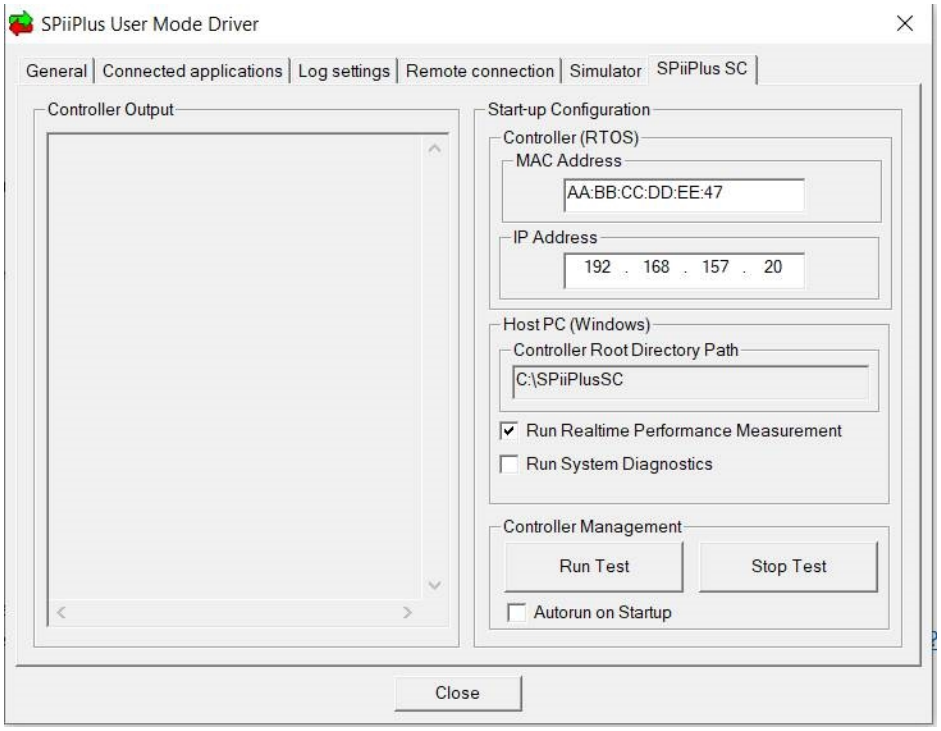
If the simulator cannot be run because either the specified ports are occupied or the simulator executable has not been set an error message is displayed.

2.7 SPiiPlusSC

The User Mode Driver has a tab for the SPiiPlusSC.

To set up a SPiiPlusSC settings:

1. Display the SPiiPlus User Mode Driver window.
2. Click the **SPiiPlusSC** tab:



This tab provides the following functionality for SPiiPlusSC:

- **Controller Output** panel - shows SPiiPlusSC controller output
- **Start-up Configurator** panel
  - **Controller (RTOS)** - Real-Time OS data
    - **MAC Address** - defined Real-Time OS MAC address
    - **IP Address** - defined Real-Time OS IP address
  - **Host PC (Windows)** - Host PC data
    - **Controller Root Directory Path** - defines SPiiPlusSC Firmware directory path on computer
    - **Username** - defines username for accessing the SPiiPlusSC Firmware directory path on Host PC (Windows) computer
    - **Password** - defines password for accessing the SPiiPlusSC Firmware directory path on Host PC (Windows) computer

The **Controller Management** panel has two action buttons:

1. **Start SC Controller** - starts the SPiiPlusSC real-time process.
2. **Stop SC Controller** - stops the SPiiPlusSC real-time process.

There is a check box for **Autorun on Startup**. When this check box is selected, the SPiiPlusSC real-time process is run automatically upon User Mode Driver start up.

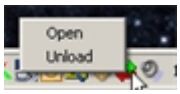
2.8 Unloading and Reloading SPiiPlus User Mode Driver

- Unloading SPiiPlus User Mode Driver
- Reloading SPiiPlus User Mode Driver

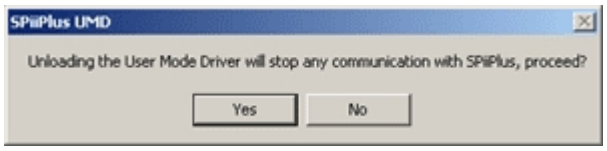
2.8.1 Unloading SPiiPlus User Mode Driver

To unload SPiiPlus User Mode Driver:

1. Right-click on the User Mode Driver icon:



2. Select **Unload**. The following prompt appears:

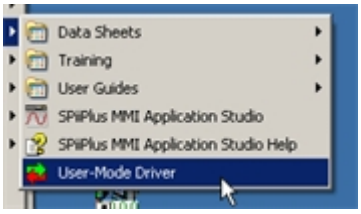


3. Click **Yes**.

2.8.2 Reloading SPiiPlus User Mode Driver

To reload SPiiPlus User Mode Driver:

1. Click **Start → Programs**.
2. Click **SPiiPlus ADK Suite vx.xx** (where x.xx represent the SPiiPlus ADKSuite installed version) and click **User-Mode Driver**. For example:



3. SPiiPlus Upgrader

3.1 Overview

Normally SPiiPlus Motion Controller firmware is upgraded using the SPiiPlus MMI Application Studio **Upgrade and Recovery** wizard. However, the option exists for upgrading the firmware using **UPGRADER.EXE** via the command line. This section provides the syntax for using this command.



Upgrading the controller firmware is a dangerous operation that may destroy the controller if executed improperly. It is strongly recommended using the SPiiPlus MMI Application Studio **Upgrade and Recovery** wizard and following all its directions carefully.

3.2 Command Line

Run the **UPGRADER** command with the path to the directory containing UPGRADER.EXE (unless the path has already been defined in a PATH statement in the AUTOEXEC.BAT file) as follows:

- Windows 64 bit:
  - SPiiPlus ADK v23.0
    C:\Program Files (x86)\ACS Motion Control\SPiiPlus ADK Suite v2.30\SPiiPlus MMI Application Studio\upgrader.exe [parameters]
  - SPiiPlusNT v.2.28 and higher:
    C:\Program Files (x86)\ACS Motion Control\SPiiPlusNT Suite x.xx\SPiiPlus MMI Application Studio\upgrader.exe [parameters]
  - SPiiPlusNT v.2.20 and lower:
    C:\Program Files (x86)\ACS Motion Control\SPiiPlus NT Suite x.xx\SPiiPlus MMI Application Studio\upgrader.exe [parameters]
- Windows 32 bit:
  - SPiiPlus ADK v23.0
    C:\Program Files\ACS Motion Control\SPiiPlus ADK Suite v2.30\SPiiPlus MMI Application Studio\upgrader.exe [parameters]
  - SPiiPlusNT v.2.28 and higher:
    C:\Program Files\ACS Motion Control\SPiiPlusNT Suite x.xx\SPiiPlus MMI Application Studio\upgrader.exe [parameters]
  - SPiiPlusNT v.2.20 and lower:
    C:\Program Files\ACS Motion Control\SPiiPlus NT Suite x.xx\SPiiPlus MMI Application Studio\upgrader.exe [parameters]

where x.xx represents the SPiiPlusNT Suite installed version

Example for Windows 32 bit with SPiiPlusNT v2.28:

C:\Program Files\ACS Motion Control\SPiiPlusNT Suite 2.28\SPiiPlus MMI Application Studio\Upgrader.exe /ETHERNET=10.0.0.100

The parameters are described in UPGRADER Parameters.

3.3 UPGRADER Parameters

The UPGRADER parameters:

- /SOURCEDIR
- /BACKUPDIR
- /<LINK TYPE>
- /DUPLICATE

3.3.1 /SOURCEDIR

This parameter defines directory containing the source files:

/SOURCEDIR=<path>

<path> - the full path to the directory that contains the firmware files to be loaded to the controller as follows:

- Windows 64 bit:
  - SPiiPlus ADk Suite v2.30:
    C:\Program Files (x86)\ACS Motion Control\SPiiPlus ADK Suite v2.30\Firmware
  - SPiiPlusNT v.2.28 and higher:
    C:\Program Files (x86)\ACS Motion Control\SPiiPlusNT Suite x.xx\Firmware
  - SPiiPlusNT v.2.20 and lower:
    C:\Program Files (x86)\ACS Motion Control\SPiiPlus NT Suite x.xx\Firmware
- Windows 32 bit:
  - SPiiPlus ADK Suite v2.30:
    C:\Program Files\ACS Motion Control\SPiiPlu ADK Suite v2.30\Firmware
  - SPiiPlusNT v.2.28 and higher:
    C:\Program Files\ACS Motion Control\SPiiPlusNT Suite x.xx\Firmware
  - SPiiPlusNT v.2.20 and lower:
    C:\Program Files\ACS Motion Control\SPiiPlus NT Suite x.xx\Firmware

where x.xx represents the SPiiPlusNT Suite installed version

If this parameter is not specified, the firmware will not be updated unless the /D switch is used.

3.3.2 /BACKUPDIR

This parameter is used for backing up all data files stored in the controller's flash into a directory on the host computer. Its format is:

/BACKUPDIR=<path>

<path> - The full path to the directory where the Upgrader will save the data files. During execution, the Upgrader creates the backup directory on the host hard disk and backs up to it the files that are replaced during the upgrade process. If this parameter is not specified, there will be no backup.

Starting from version NT 2.1, the functionality of the **/BACKUPDIR** parameter has been expanded to backup the following data:

- Firmware

> SP Files

> EtherCAT Configuration File

The EtherCat configuration information is stored in the the controller's flash. The Upgrader checks for the existence of the EtherCAT configuration file in the controller's flash, and, if it exists, the system copies it to the backup folder. This file is required for /DUPLICATE further operation.

> System Configuration File

The system configuration information is stored in the controller's flash. The Upgrader checks for the existence of the system configuration file in the controller flash, and, if it exists, copies it to the backup folder. This file is required for further /DUPLICATE operation.

> Application File

The application data consists of system and axis parameters that are stored in the controller's flash. A new argument has been added to the Upgrader utility for use with **/BACKUPDIR**:

/BACKUPAPPLNAME=<applfilename>

When used, an application file, **<appfilename>.spi**, is created and stored in the backup folder. If **/BACKUPAPPLNAME** is not specified, then the default name: **Backup.spi** is given to the application file. This file is required for further /DUPLICATE operation.



The <applfilename> argument should match the Windows operation system file names convention, otherwise the Upgrader utility will stop execution and inform the user about the problem.

> Checksum File

The backup folder is locked using a folder with an integrity file: **upgreg.###** (which is similar to the firmware upgrade folder). The **upgreg.###** file is necessary for the /DUPLICATE operation.



Not all earlier firmware versions support backup for all the communication channels. Therefore, depending on the communication channel and firmware version, the backup option may not available.

3.3.3 /<LINK TYPE>

Communication link information. This parameter is obligatory. The format is:

/<LINK TYPE>=<par1,[par2]>

<LINK TYPE> – connection type: SERIAL, ETHERNET or PCI.

<par1> – COM port number if <LINK TYPE> is SERIAL; or TCP/IP address if <LINK TYPE> is ETHERNET Slot number (or -1 for autodetection) if <Link Type> is PCI

<par2> – Baud rate if <LINK TYPE> is SERIAL

Examples

UPGRADER.EXE /SERIAL=1,115200 – Serial connection and baud rate.

UPGRADER.EXE /ETHERNET=10.0.0.100 – Ethernet connection and address.

UPGRADER.EXE /PCI= -1 – PCI connection with autodetection.

3.3.4 /DUPLICATE

A new Upgrader command line parameter, **/DUPLICATE**, has been added to perform a Duplicate Machine operation. This operation duplicates the files of one machine on another machine.

The operation ensures that all necessary files exist. It also checks the source folder integrity and will stop the execution in case of integrity test failure.



/DUPLICATE can only be used if /BACKUPDIRhas been previously performed. The regular upgrade folder which was not created using **/BACKUPDIR** cannot be used for duplicate operation. However, a selective files upgrade is still available if **/DUPLICATE** is not used.

The format of the parameter is:
/DUPLICATE arguments

The following arguments are required:

- /SOURCEDIR
- /<LINK TYPE>

3.4 Switches

The switches which may be employed when running Upgrader:

- /D
- /NB
- /Y

3.4.1 /D

Causes Upgrader to load the firmware from the directory specified by /BACKUPDIR (the backup directory). In this case the **/SOURCEDIR** argument is not required.

3.4.2 /NB

Indicates that no backup is required.

3.4.3 /Y

Automatic confirmation to all questions that the Upgrader might pose during the upgrade process.

3.5 Application Arguments

3.5.1 /APPLFILE

This argument refers to an application file that has been saved using the Application Saver feature in SPiiPlus MMI Application Studio. Its format is:
/APPLFILE=<path>

<path> - The full path to the directory where the Upgrader will find the file.

If this argument is not specified, the other application-related arguments (/APPLxxx) will not be processed.

3.5.2 /APPLBUF

This argument specifies the controller buffers to which the ACSPL+ programs in the /APPLFILE will be loaded. (There must be an ACSPL+ program in the corresponding /APPLFILE buffer.) Its format is:
/APPLBUF=<buffer_enum>

<buffer_enum> – A comma-separated list of buffer numbers (starting from zero), or ALL for all buffers, or NONE to prevent the loading of any application at all.

If this argument is not specified, the default value ALL is assumed.

Examples:

UPGRADER.EXE /APPLBUF=0,1,2 – Loads buffers 0 through 2.

UPGRADER.EXE /APPLBUF=0,1,2,10 – Loads buffers 0 through 2 and buffer 10.

UPGRADER.EXE /APPLBUF=ALL – Loads all buffers, including the D-Buffer.

UPGRADER.EXE /APPLBUF=NONE – Does not load any buffers.

3.5.3 /APPLPAR

This argument specifies the controller configuration parameters that can be loaded to the controller from the /APPLFILE. (This assumes that the corresponding axis (and/or system) configuration parameters have been saved in the /APPLFILE.) Its format is:
/APPLPAR=<axis_enum>

<axis_enum> – A comma-separated list of axes (0, 2, 3, etc.) and/or system (S), or ALL for all configuration parameters, or NONE to prevent to prevent any configuration parameters from loading.

If this argument is not specified, the default value ALL is assumed.

Examples:

UPGRADER.EXE /APPLPAR=S, 0, 1, 2, 4 – Loads saved system configuration parameters and saved configuration parameters for axes 0, 1, 2, and 4.

UPGRADER.EXE /APPLPAR=ALL – Loads all saved axis and system configuration parameters

UPGRADER.EXE /APPLPAR=NONE – Prevents any configuration parameters from being loaded.

3.5.4 /APPLADJ

This argument specifies the adjustment parameters that can be loaded to the controller from the /APPLFILE. (This assumes that the corresponding axis adjustment parameters have been saved in the /APPLFILE.) Its format is:
/APPLADJ =<axis_enum>

<axis_enum> – A comma-separated list of axes (0, 2, 3, etc.), or ALL for all the adjustment parameters, or NONE to prevent any adjustment parameters from loading.

If this argument is not specified, the default value ALL is assumed.

Examples:

UPGRADER.EXE /APPLADJ=0, 1, 2, 4 – Loads saved adjustment parameters for axes 0, 1, 2, and 4.

UPGRADER.EXE /APPLADJ=ALL – Loads all saved adjustment parameters.

UPGRADER.EXE /APPLADJ=NONE – Prevents any adjustment parameters from being loaded.

3.5.5 /APPLSP

This argument specifies the SP program numbers that can be loaded to the controller from the /APPLFILE. (This assumes that the corresponding SP programs have been saved in the /APPLFILE.) Its format is:
/APPLSP=<SP_enum>

<axis_enum> – A comma-separated list of SP processor numbers (starting from zero), or ALL for all the SP processors, or NONE to prevent any SP programs from loading.

If this argument is not specified, the default value ALL is assumed.

Examples:

UPGRADER.EXE /APPLSP=0, 1, 2, 3 – Loads the programs from SPs 0 to 3.

UPGRADER.EXE /APPLSP=ALL – Loads all the SP programs.

UPGRADER.EXE /APPLSP=NONE – Prevents any SP programs from being loaded.

3.5.6 /APPLUSERFILE

This argument specifies the user files that can be loaded to the controller from the /APPLFILE. (This assumes that the corresponding user files have been saved in the /APPLFILE.) This argument can be used to load arrays of user or standard variables by first saving them to the controller using the WRITE command. Its format is:
/APPLUSERFILE=<file_enum>

<axis_enum> – a comma-separated list of user file names, or ALL for all user files, or NONE to prevent any user files from loading.

If this argument is not specified, the default value ALL is assumed.

Examples:

UPGRADER.EXE /APPLUSERFILE=myfile1,myfile2

UPGRADER.EXE /APPLUSERFILE=ALL

UPGRADER.EXE /APPLUSERFILE=NONE

4. SPiiPlus Simulator

4.1 About the SPiiPlus Simulator

The SPiiPlus Simulator is a Windows application that simulates the real controller behavior. The Simulator is a powerful tool for use during application development and debugging.

The SPiiPlus Simulator allows you to work with the SPiiPlus MMI Application Studio, SPiiPlus C Library, SPiiPlus COM Library, or SPiiPlus .NET Library without being physically connected to a controller, drives, or motors. The Simulator program simulates the feedback from a SPiiPlus controller connected to drives and/or motors with no actual hardware involved. In this way it provides a way to develop your motion programs before connecting to actual hardware.

The Simulator emulates all programming features of the controller, **except** for the following:

- Feedback from real motors: Instead the simulator sets the feedback values equal to the reference values, which is equivalent to ideal motors with zero following error. This means the feedback position and reference position are equal (or, in terms of the associated ACSPL+ variables: **FPOS**=**RPOS**).
- Real time operation: The simulator emulates the controller time and supplies the **TIME** variable, however **TIME=1** is not equal to 1 millisecond as when working with a real SPiiPlus.
- Some hardware-based programming features, including: Index, **MARK**, **PEG**, and analog I/O (**AIN** and **AOUT**).
- Some SPiiPlus MMI Application Studio **Communication Terminal** commands: **#HWRES**, **#RESET**, **#U**, **#IR, #SI**, **#SIR**.
- All **write/read** commands.
- The SPiiPlus MMI Application Studio **Application Saver** and **Application Loader**.



The Simulator does, however, provide limited support for SPiiPlus MMI Application Studio **Adjuster**.

4.2 Running the Simulator

There are four ways to run to the Simulator:

1. The SPiiPlus MMI Application Studio
2. The SPiiPlus C Library, SPiiPlus COM Library, or SPiiPlus .NET Library
3. The SPiiPlus User Mode Driver
4. The computer command line

4.2.1 Using the SPiiPlus MMI Application Studio

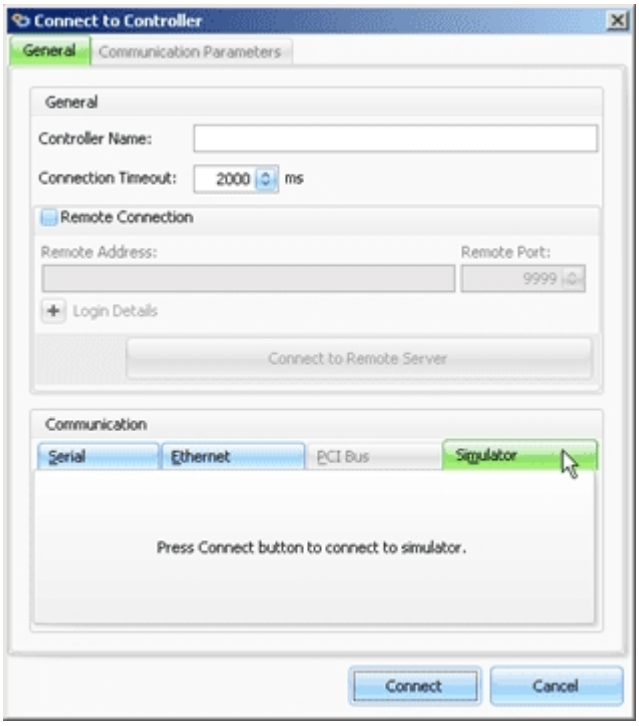
To activate the Simulator through the SPiiPlus MMI Application Studio:

1. Click **Workspace** on SPiiPlus MMI Application Studio Menu bar, and select **Add Controller**.



The Connect to Controller dialog is displayed.

2. Click **Simulator**.



3. Click **Connect**. The "controller" is added to the Workspace Tree. The controller is, in fact, the Simulator.

4.2.2 Using the SPiiPlus C Library, SPiiPlus COM Library, or SPiiPlus .NET Library

The Simulator launches automatically with any attempt to connect to it using the SPiiPlus C Library, SPiiPlus COM Library or the SPiiPlus .NET Library.

Connecting to the Simulator

Enter the appropriate command as shown below.

Command	Interface
acsc_OpenCommSimulator	SPiiPlus C Library function
OpenCommSimulator	SPiiPlus COM Library method
OpenCommSimulator	SPiiPlus .NET Library method

For details, refer to the *SPiiPlus C Library Programmer's Guide*, *SPiiPlus COM Library Programmer's Guide*, or the *SPiiPlus .NET Library Programmer's Guide*.

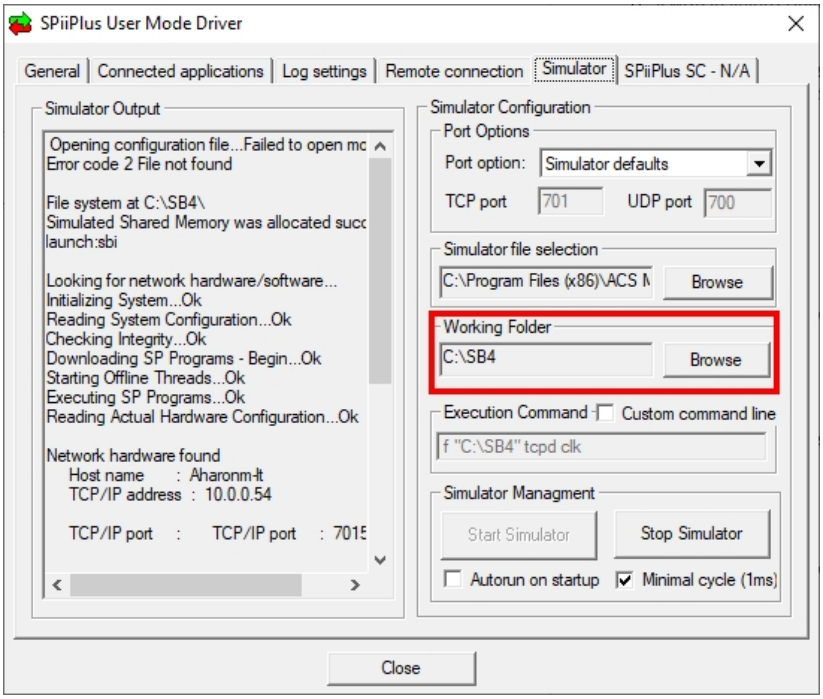
4.2.3 Using the SPiiPlus User Mode Driver

Activate the Simulator using the SPiiPlus User Mode Driver:

1. Display the SPiiPlus User Mode Driver window, see To View the SPiiPlus User Mode Driver.
2. Click the **Simulator** tab.

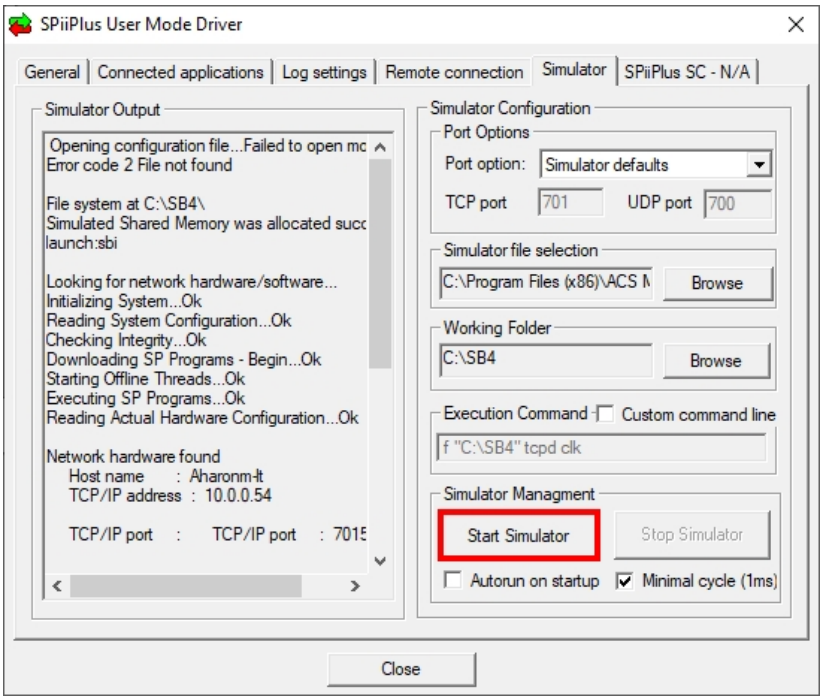
Starting the SPiiPlus simulator:

> To set an alternative working folder, use the Working Folder text field in the Simulator Tab.



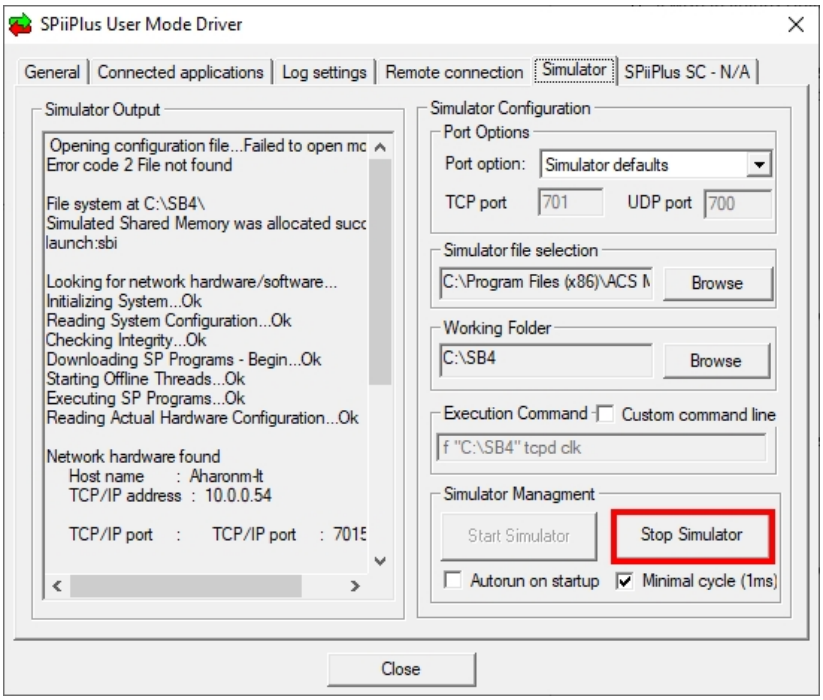
Folders requiring user permissions cannot be used as working folders. For example, if a folder under "C:\users\..." is set as the working folder the simulator will not work correctly.

> Press **Start Simulator** in the Simulator Tab.



Stopping the SPiiPlus simulator:

> Press **Stop Simulator** in the Simulator TAB.



4.2.4 Using the Command Line



Running the simulator using the command line is a legacy application. It uses a different process than the SPiiPlus Simulator. This is not a recommended operation.



After running the simulator using the command line it is not possible to connect to the simulator using the simulator connection functionality available within the SPiiPlus MMI Application Studio, SPiiPlus C Library or SPiiPlus COM Library. A TCP connection must be used.

To run the Simulator using the command line:

1. On drive C:\ create the folder: **SB4**.
2. Execute the Simulator by selecting **Start → Run** and entering the Simulator activation command with the latest Simulator path:
   - Windows 64 bit:
     - SPiiPlus ADK Suite v2.30:
       C:\Program Files (x86)\ACS Motion Control\SPiiPlus ADK Suite v2.30\User Mode Driver\sb4.exe
     - SPiiPlusNT v.2.28 and higher:
       C:\Program Files (x86)\ACS Motion Control\SPiiPlusNT Suite x.xx\User Mode Driver\sb4.exe
     - SPiiPlusNT v.2.20 and lower:
       C:\Program Files (x86)\ACS Motion Control\SPiiPlus NT Suite x.xx\User Mode Driver\sb4.exe
   - Windows 32 bit:
     - SPiiPlus ADk Suite v2.30:
       C:\Program Files\ACS Motion Control\SPiiPlus ADK Suite v2.30\User Mode Driver\sb4.exe
     - SPiiPlusNT v.2.28 and higher:
       C:\Program Files\ACS Motion Control\SPiiPlusNT Suite x.xx\User Mode Driver\sb4.exe
     - SPiiPlusNT v.2.20 and lower:
       C:\Program Files\ACS Motion Control\SPiiPlus NT Suite x.xx\User Mode Driver\sb4.exe

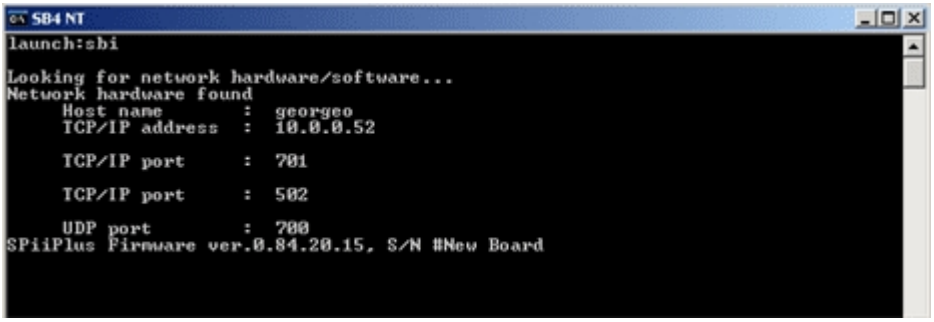
   where x.xx represent the SPiiPlusNT Suite installed version.

Example for Windows 32 bit with SPiiPlusNT v2.28:

"C:\Program Files\ACS Motion Control\SPiiPlusNT Suite 2.28\User Mode Driver\sb4.exe" f C:\SB4 tcpd clk

where the **f** tag is the program's ID.

3. The simulator finds the network settings of the host and displays them on the console.



4.2.4.1 The TCP Parameters

The following four parameters can be used for TCP initialization:

> tcp

If **tcp** is specified, the Simulator activates the network and opens the UDP port and TCP port with the address supplied by Windows.

> tcpg

If **tcpg** is specified, the Simulator activates the network and opens the UDP port and TCP port with addresses supplied by WINDOWS and writes the port numbers in the registry key "HKEY_LOCAL_MACHINE\SOFTWARE\ACS-Tech80\SPiiPlus Simulator" as values "UDP port" and "TCP port".

> tcpd

If **tcpd** is specified, the Simulator activates the network and opens fixed ports: UDP port 700 and TCP port 701.

> t<tcp_port_number>_u<udp_port_number>

If **t<tcp_port_number>_u<udp_port_number>** is used, the Simulator activates the network and opens the TCP and UDP ports specified by **tcp_port_number** and **upd_port_number**.

4.2.4.2 The clk Parameter

In this case the Simulator should be launched manually using the following command:

sb4.exe f C:\SB4 tcpd clk.

The **clk** parameter forces the Simulator in all cases to run with a 1ms cycle time by setting the “tick” period of Windows to this minimum (the default value is 10ms). This ensures that all applications run at a reasonable speed.

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